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A. 1) Research Program/Project/Study Title: Research Title: Comparative Study of Powdered and Extracted condiments (Batwan, Libas, Labog, and Kamias) in Deboned Bangus	
Proponent Name:	JANET V. LEDESMA
Academic Rank:	Instructor I
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	DAYAO SATELLITE COLLEGE
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz

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Region: VI	
(5) Classification Research: ☒ Basic ☐ Applied Promotion/Commercialization Development: ☐ Pilot Testing ☐ Tech. 	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity	(8) Discipline/Field of Specialization Food Production
(9) Rationale: The researcher came up with the idea of Comparing powder and extract using Batwan, Libas, Labog, and kamias. Filipinos, especially their ancestors, were likely to use souring agents around them, which stayed on for succeeding generations in the Philippines. They are used in any food that involves souring agents' flavors like pork, beef, and even in chicken. However, some Filipinos are not familiar with the souring agents' flavors around them they don't know what is the purpose of those souring agents. The four souring agents flavor mentioned above have the nutrition wherein need in the body, the four souring agents have a good source of vitamins C and A, improve calcium absorption, antimicrobial, anti-diabetic, and ulcer-protective. In this study, the researcher aims to produce organic indigenous flavors such as Kamias, Libas, Batwan, and Labog that can apply any food like boneless Bangus for additional flavor and curing agent. There are existing souring flavor powders in the market which as the sinigang mix, which is common for Filipinos. But for this study, the researcher wants to know if this study can be an acceptable souring agent and have the potential to commercialize and determine the acceptable souring flavors for any food. The researchers also aim to develop a value-added product for the indigenous sour agents in the market.	
(9-A) Study Description This study aims to determine the comparison of powder and extraction between batwan, libas, labog, and kamias for consumers as a condiment additional flavored. The result of this study will serve as proof that the comparative of powdered and extracted condiments between Batwan, Libas, Labog and kamias has potential for commercialization.	
(10) Objectives/Statement of the Problem A. Objectives: This study will be conducted to determine the acceptability between the powdered and extracted condiment of an indigenous souring agent such as batwan, libas, labog, and kamias. <i>Specifically, it aims to</i> Determine the acceptability of the souring level of four treatments such as	

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- batwan,libas,labog, and kamias.
2. To determine the most acceptable method between powder and extract.
3. To determine the acceptability in terms of taste, odor, color, and texture of the products.
4. To determine the cost analysis.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology/information from which the project proposal will take off).

This study relates to the additional flavoring condiments for any kind of food that need souring agents.

Batwan, libas,labog, and kamias are known as souring agents and have quite nutritious and rich in plant compounds called antioxidants, which help protect your cells from damage and also prolong food spoilage. These souring agents are indigenous and Capiz is rich in these flavors. The process of making powder flavor is the following steps: for batwan and kamias, Washing, cutting/chopping to remove the seeds, dehydrating, and blending until powder. for the libas and labog leaves, Washing, dehydrate and blend until powder. For the process of making extract: for batwan and kamias, Washing, cutting/chopping to remove the seeds, blend, and squeeze. For the libas and labog leaves, Washing, blending and squeezing.

Process of Boneless Bangus

The Bangus will be washed, and split fish into butterfly fillets. Remove the internal organs. Wash thoroughly. Remove the backbone and dorsal fin. Make a line with the use of forceps, start removing the rib bones, and make a slit along with the vent of the dorsal and ventral muscles with the edge of a knife. Remove the spines one after the other by inserting the pair of forceps between the segments of the dorsal and ventral muscles. Brine the boneless bangus with water, salt, and sugar solution for 30 minutes. After 30 minutes of brining, remove the boneless Bangus from brining solution. Marinate the fish with four processed both powdered and extracted condiments batwan, libas, labog, and kamias for (2 hours). Set aside.

(12) Scientific Basis/Theoretical Framework (Include concepts. Theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experiment procedure and design; d) statistical analysis; e) evaluation, method, and observations to be made, strategies for implementation.)

Materials and Methods

The experimental studies will be conducted at Capsu Dayao Fisheries and Food Research Development Center (FFRDC).

Materials:

Knife
Scissor
Dehydrator
Measuring spoon
Weighing Scale
Mixing bowl
Colander
Blender

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Experimental Treatment

	Treatment A (Batwan)	Treatment B (Libas)	Treatment C (Labog)	Treatment D (Kamias)
Powdered	50g	50g	50g	50g
Extracted	50g	50g	50g	50g
Bangus	500g	500g	500g	500g

Procurement of raw material

Raw materials for this study are batwan, libas, labog and kamias purchased from the local market Bagong Lipunan in Roxas City.

Preparation of Boneless Bangus

Washing Bangus
 ↓
 Splitting
 ↓
 Cleaning
 ↓
 Deboning
 ↓
 Weighing

Preparation of batwan and kamias in powdered processing

Washing
 ↓
 Cut/chop
 ↓
 Dehydrating
 ↓
 Blending until powderize

Preparation of libas and labog in powdered processing

Washing
 ↓
 Dehydrating
 ↓
 Blending until powderize

Preparation of batwan and kamias in extracted processing

Washing
 ↓
 Cut/chop
 ↓
 Blending
 ↓
 Squeezing

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Preparation of libas and labog in extracted processing

Washing
 ↓
 Blending
 ↓
 Squeezing

Sensory Evaluation of Dried Flavored Boneless Bangus

The sensory evaluation consists of judging the quality of condiments by a panel of 10 subject experts, and 100 consumers. The evaluation deals with analyzing and interpreting the qualities of the powdered and extracted condiments (batwan, libas, labog and kamias) applied to fresh boneless bangus. The 9-point Hedonic scale score-card method was used to determine the sensory characteristics of dried flavored boneless Bangus. Sensory attributes of the powder and extract condiments (batwan, libas, labog and kamias) applied in fresh boneless bangus are appearance, texture, taste, aroma, and overall acceptability will be allotted a maximum of 9 each. The scoring scale has a rating of 1 (dislike extremely), 2 (dislike very much), 3 (dislike moderately), 4 (dislike slightly), 5 (neither like or dislike), 6 (liked slightly), 7 (liked moderately), 8 (liked very much), 9 (liked extremely).

Experimental Design and Treatment

Experimental Design

The study uses the experimental method of research. An experimental method is a method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to the mixture of the different samples (Calderon, 2002).

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

	1 st Quarter	2 nd Quarter	3 Quarter
Research Objectives			
Preparation of proposal and other related activities			
Experimental plan			
Preparation of bangus and other materials			
Sensory Analysis			
Preparation and writing of data and results			

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Presentation of Results																																																											
(15) Expected Output (Indicate the specific products, processes, or services that the project is expected to produce; quantify when possible) To produce indigenous Cured boneless bangus for consumers in Roxas City, in the Province of Capiz and neighboring provinces.																																																											
(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes, or services to be transferred as extension activities) The process will be taught to the target clientele in the community through the extension program of the college. Present the technology to the student. Present the study in a research symposium. Commercialization of Cured boneless bangus for family consumption and or backyard micro industry.																																																											
(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the project's output) Cured boneless bangus for family consumption, entrepreneur, retailers, restaurants, students, faculty, staff, and/or for backyard micro-industry.																																																											
(17) Personnel Requirement <table border="1"> <thead> <tr> <th>Name</th> <th>Role (Program,Project, Study Leader)</th> <th>Field of Specialization</th> </tr> </thead> <tbody> <tr> <td>Janet V. Ledesma</td> <td></td> <td>Food and Beverage</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> </tr> </tbody> </table>				Name	Role (Program,Project, Study Leader)	Field of Specialization	Janet V. Ledesma		Food and Beverage																																																		
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(19) Literature Cited (include only those references published in reputable journals, and at least that from)

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<http://pinoytrees.blogspot.com/2009/09/souring-agents-of-sinigang.html>

<https://www.pinoybisnes.com/food-business-ideas/bangus-de-boning-boneless-bangus/>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

JANET V. LEDESMA

Villa Alegria Subdivision

Punta Tabuc, Roxas City

Capiz, Philippines, 5800

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Work Experience:

Hercor College: Faculty 2005-2015

General Manager : Hercor Catering Services 2005-2015

Subject Handle: Asian Cuisine, International Cuisine, Food and Beverage, Catering Management, Costing Management, Bakery Management, Total Quality Management, Bread and Pastry.

Capiz State University: Instructor 2015 to present

Subject Handle: Bakery Management, Food Safety, Principles to Human Nutrition, Entrepreneurship, Trade and Marketing

Gabriel's Catering and Services-owner 2015 to present

Gabriel's Event Center-owner

Rafael's Cakes-owner

Village Coffee-Owner

Academic history

Doctoral Central Philippine University 2021
Jaro, Iloilo

Doctor of Tourism and Hospitality Management

Master's Colegio de la Purisima Concepcion 2010

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Tertiary	Roxas City, Capiz	
	Master of Business Administration (M.B.A.)	
	Colegio Del Sagrado Corazon de Jesus	1996
	Iloilo City	
	Bachelor of Science in Hotel and Restaurant Management	
Licenses and certifications.		
• NC II Holder in Bread and Pastry Production • NC II Holder in Food and Beverage Services • NC II Holder in Cookery • Fundamentals of Culinary Arts (International School for Culinary Arts & Hotel Management) • Bread and Pastry Production NII Training Program (St. Dominic College of Asia)		